Multiscale Gaussian Attention Mechanism for Tiny-Object Detection in Remote Sensing Images

Shuohao Shi, Qiang Fang, Xin Xu, *Senior Member, IEEE*, and Dezun Dong

***Abstract*—Tiny-object detection is increasingly crucial in fields such as remote sensing, traffic monitoring, and robotics. Inspired by human visual perception, the attention mechanism has become a widely used method for enhancing object detection performance. While existing attention mechanisms have significantly advanced general object detection performance, they often fall short in adapting to the characteristics of tiny-object datasets, including huge object size variations and concentrated distributions. In detail, most current attention mechanisms rely on convolutional or linear layers with fixed receptive fields to compute attention vectors. Some methods attempt to enlarge the receptive fields by using multiscale structures, but they often simply sum feature maps, leading to information interference and increased computational costs. To address these issues, we propose a novel multiscale Gaussian attention mechanism (MGAM). This mechanism integrates multiscale receptive fields with dynamic feature weighting and a Gaussian attention module (GAM), replacing traditional convolutional layers to reduce training and inference overhead. In addition, our mechanism can be easily embedded into various detectors without any hyperparameters. Extensive experiments on six object detection datasets demonstrate the effectiveness and robustness of our method. The code is available at https://github.com/cszzshi/MGAM**

***Index Terms*—Attention mechanism, deep learning, object detection, tiny-object detection.**

I. Introduction

With the rapid development of artificial intelligence, which has been applied in various fields, such as spectral big data application [1], radar scattering processing [2], millimeter-wave imaging [3], medical diagnosis [4], and road segmentation [5], object detection is one of the most widely used artificial intelligence technologies in our daily lives, such as intelligent transportation and environmental protection. Particularly, driven by the widespread adoption of deep learning and the introduction of the ResNet [6], the performance of object detection has seen significant improvements in both speed and accuracy. For example, the latest version of YOLO, YOLOv10 [7], achieves millisecond-level detection with competitive accuracy while utilizing only 2.3 M parameters.

Received 8 May 2025; accepted 17 July 2025. Date of publication 21 July 2025; date of current version 5 August 2025. *(Corresponding author: Qiang Fang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3591260

Recently, many researchers have focused on enhancing the performance of object detectors. There are a series of latest proposed methods across various fields, such as the StarNet [10], DQ-DETR [11], and RegSeg [12], especially in the field of remote sensing object detection, for example, the GLSANet [13] reduces the interference of background noise on objects by mining and extracting deep features in the network and designs an optimized feature pyramid network (FPN) to enhance the multiscale detection capability in remote sensing images. To alleviate the problem of insufficient labeled data in remote sensing object detection tasks, Wang et al. [14] proposed a multilevel information exchange strategy and a semantic enhancement module to improve the performance of few-shot remote sensing object detection. From the perspective of knowledge distillation, the APDNet [15] and ACFG [16] have also achieved satisfactory results in the field of remote sensing image object detection. In addition, attention mechanisms inspired by the observation manner of human beings are one of the most popular research fields. Considering the significance of each channel in a feature map may be different, channel attention mechanisms have been proposed. For example, the SENet [17] and ECA-Net [18] employ a fully connected layer and 1-D convolutional layer, respectively, to compute channel attention vectors, which are then multiplied by the original feature map to produce the final output. The SKNet [19] fuses features from different kernel sizes before computing the channel attention vector. Similarly, since the contribution of various locations in a feature map may differ, spatial attention mechanisms have been introduced. For example, the DCN [20] proposed deformable convolution, allowing the network to automatically select appropriate receptive fields based on spatial locations. Some methods combine both channel and spatial attention, for example, CBAM [21] employs a cascade structure to sequentially compute channel attention and spatial attention vectors.

Considering the lack of information and relatively concentrated distributions, the attention mechanism is also crucial for improving tiny-object detection performance. However, through our observation and analysis, there are mainly three aspects of problems that may occur if we directly apply existing attention mechanisms to tiny-object detection tasks.

First, when computing channel and spatial attention, some methods rely on a single kernel size for convolution, which provides only one type of receptive field. However, object sizes in tiny-object detection datasets can vary significantly. For example, the VisDrone2019 [22] dataset contains both very

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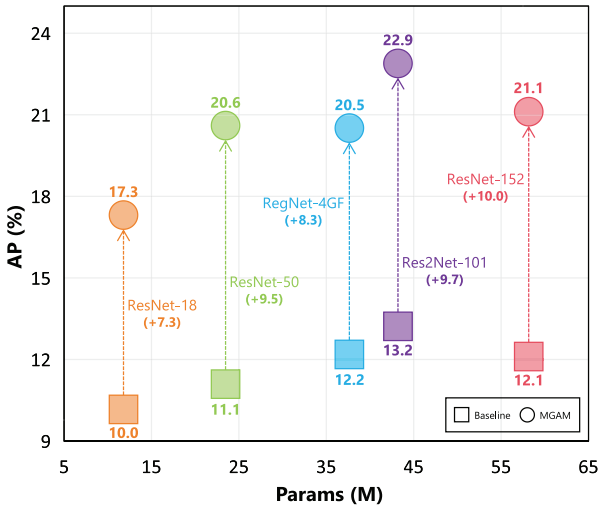


Fig. 1. Performance comparisons between the baseline and our MGAM on different backbones, including the ResNet [6], RegNet [8], and Res2Net [9]. All results are based on faster R-CNN and on the tiny-object detection dataset AI-TOD. It is obvious that our MGAM not only has strong robustness for different backbones and depths but also does not introduce any extra parameter.

tiny and large objects within the same image. Consequently, a fixed receptive field may not be suitable for scenes with varying object scales. While some improved methods utilize multiple kernel sizes during computation, they often simply aggregate them without considering that different receptive fields may carry different weights.

Second, current approaches often treat channel and spatial attention separately, neglecting cross-domain information integration. Although some methods do consider both channel and spatial attention, they often employ a cascade structure or simply multiply the two types of attention features, which may lead to information confusion during training.

Third, most existing channel and spatial attention mechanisms depend on a series of convolutional or fully connected layers, which can introduce additional parameters to the network. For example, after using the attention mechanism of CBAM [21] in backbone ResNet-50 [6], the parameters will increase from 23.5 to 29.1 M. Moreover, most of the tiny objects within an image are often concentrated in a specific region, yet many current methods for computing attention features do not account for this. As a result, these methods can be easily disrupted by surrounding noise, preventing the model from effectively focusing on the most important channels or regions.

To address the aforementioned problems and further improve the tiny-object detection performance, we propose a novel attention mechanism specifically designed for tiny-object detection, named the multiscale Gaussian attention mechanism (MGAM). This mechanism simultaneously considers both channel and spatial attention and leverages information from different receptive fields through a specially designed dynamic weighting module (DWM). Additionally, our mechanism can be seamlessly integrated into mainstream backbones, leading to significant improvements in detection accuracy, as illustrated in Fig. 1. The main contributions of this article are as follows.

1) We propose a multiscale feature extraction module (MFEM) at the beginning of our attention mechanism to extract features with varying receptive fields, thereby accommodating object scale variations.
2) We present a novel Gaussian attention module (GAM) to compute the attention features, which can make the model better focus on important channels and locations.
3) We design a DWM that dynamically assigns weights to each feature map from different receptive fields, fully leveraging multiscale information while reducing interference.
4) We integrate the above three modules to form the complete structure of the MGAM. Extensive experiments demonstrate the excellent performance of our method, which can be easily embedded into various backbones.

The rest of the article is organized as follows. In Section II, we make a brief review of the related work for tiny-object detection, attention mechanisms, and feature fusion. In Section III, we provide a detailed introduction to our proposed method. In Section IV, we conduct extensive experiments on both tiny- and general-object detection datasets to prove the effectiveness of our method. In Section V, we conclude the whole article.

II. Related Work

A. Tiny-Object Detection

Tiny-object detection is increasingly recognized as one of the most challenging tasks in computer vision. Tiny objects usually refer to objects with an absolute size less than 32 pixels, which is defined in MS COCO 2017 [23]. According to a comprehensive review on tiny-object detection [24], the extremely small size and low relative area of tiny objects lead to significant challenges, including information loss, interference from noisy information, and insufficient training samples, all of which contribute to a substantial decline in detection performance compared to general object detection. Recently, many researchers have focused on improving the accuracy and speed of tiny-object detection. One approach to addressing these challenges involves increasing the number of training samples. For example, Bosquet et al. [25] proposed a novel data augmentation method based on generative adversarial networks (GANs) to generate more tiny objects from corresponding larger ones. Another strategy to enhance training involves using more suitable label assignment techniques. For example, DotD [26] analyzed the high sensitivity of the traditional IoU metric for tiny objects and proposed Dot Distance to assign more positive samples to tiny objects, NWD [27] and Gaussian assignment (GA) [28] introduced 2-D Gaussian distribution to model each ground truth and anchor to increase the number of high-quality positive samples. To mitigate information loss, methods like the SINet [29] and FS-SSD [30] fully leverage contextual information to enhance detection performance.

B. Attention Mechanism

In daily life, when observing a scene, we often unconsciously focus on specific areas that may feature bright colors,

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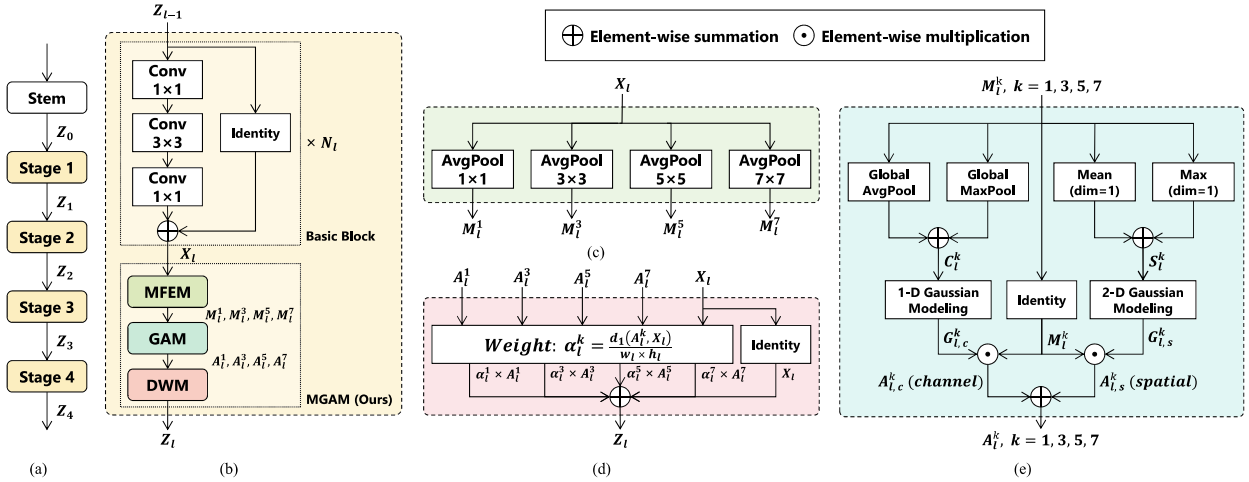


Fig. 2. Structure of the MGAM. (a) Backbone consists of four stages connected in series. (b) Each Stage l contains two parts, a basic block repeated for N_l times and an MGAM, which is our proposed module and contains three modules. (c) MFEM is used to extract features from multiscale receptive fields. (d) DWM can automatically assign different weights based on the importance of each feature. (e) GAM considers both channel and spatial attention mechanisms without introducing extra parameters. Our MGAM can be easily embedded into various backbones; the basic block and N_l are different for different backbones.

distinct markings, or clear contrasts with their surroundings. Inspired by this natural human tendency, researchers have introduced various attention mechanisms designed to enable models to automatically focus on the most important parts of an image. These mechanisms have been widely applied across multiple fields, including machine translation, speech recognition, and so on. Recent computer vision studies classify object detection attention mechanisms into three categories [31]: channel-based approaches (SKNet [19], SENet [17], ECA-Net [18]), spatial-based approaches (DCN [20]), and hybrid channel–spatial frameworks like CBAM [21].

Despite significant progress having been made in developing attention mechanisms for general object detection, there are relatively few mechanisms specifically designed for tiny-object detection, particularly for images captured by uncrewed aerial vehicle (UAV) and satellites. Existing methods in this area include the SCRDet [32], DAM [33], ABP [34], ECA [35], ARPN [36], and ESG_TODNet [37]. The SCRDet and DAM compute both channel and spatial attention from the original feature map using a series of fully connected layers and convolutional layers and then multiply these attention maps with the original feature map to generate the output. ABP incorporates attention mechanisms into a feature pyramid, using a level-based attention module and a spatial attention module to weight each feature map and guide the model to focus on important regions. ECA employs global average pooling and fully connected layers to derive the channel attention vector, which is then multiplied by the original feature map to emphasize the important channels. ARPN designed a series of front and back attention modules to adaptively generate high-quality proposal regions for two-stage object detectors. The latest published method, ESG_TODNet, introduced an explicit semantic guidance module to generate an attention map that can emphasize the regions containing tiny objects.

C. Feature Fusion

Feature fusion empowers deep learning models by strategically combining multilevel data representations, effectively processing complex patterns to significantly boost accuracy in object detection and semantic segmentation tasks. One of the most widely adopted feature fusion architectures is the FPN [38], which utilizes a top-down pyramid structure to merge features from different scales. Based on the FPN structure, the PANet [39] introduced a set of bottom-up feature fusion pathways and adaptive feature pooling to better aggregate multiscale features. The HRNet [40] designed a repeated multiresolution fusion structure to fuse the features from different resolutions and then obtain more comprehensive information.

In addition, there are also specialized approaches designed for tiny-object detection. For example, the MFR-CNN [41] introduced a multiscale feature representation module that fuses local and global information, which is then used to predict object locations and categories. To mitigate the negative effects of the top-down connections in the FPN structure, Gong et al. [42] proposed an efficient fusion factor that adaptively controls the information flow from the top levels to the bottom levels. Besides, some latest published methods, like the HANet [43], designed a scale-specific feature re-coupling module to fuse the features from different groups and then enhance the feature space. The ORFENet [44] introduced a multiple feature enhancement module before classification and regression to enhance the features for tiny objects adaptively.

III. Method

In this section, we first provide an overview of the structure of our detection network. Following this, we will provide a detailed introduction to the three proposed modules. Finally, we will discuss how to integrate our mechanism into existing backbone networks.

A. Overview

The structure of the MGAM is illustrated in Fig. 2. As depicted in Fig. 2(a), most existing backbones are composed of a series of stages, where the output from one stage serves not only as the input for the next stage, but is also utilized

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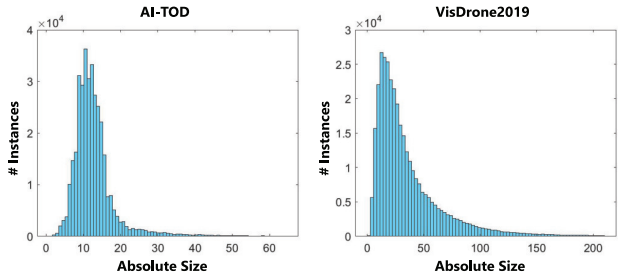


Fig. 3. Distributions between absolute size and corresponding number of instances on the AI-TOD [45] and VisDrone2019 [22] datasets.

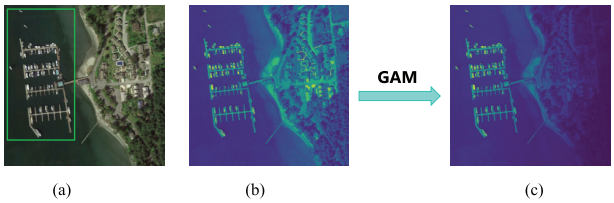


Fig. 4. Visual comparison between the original feature map and the modeled feature map after the process of the GAM. The green rectangle represents the region where objects appear. It is obvious that the useless information can be effectively suppressed by our GAM. (a) Original image. (b) Feature map. (c) Modeled feature map.

in subsequent classification and detection tasks, so we embed our MGAM into each stage of the backbone. In Fig. 2(b), we divide each stage into two parts: the basic block, which follows the structure of the base backbone, and the MGAM, which is our proposed MGAM. For example, if ResNet-50 [6] is selected as the base backbone, the basic block consists of a series of residual blocks, with the number of blocks N_l for each stage being 3, 4, 6, and 3, respectively. The output feature X_l from the basic block is fed into the MGAM, where it passes through the three modules: the MFEM, the GAM, and the DWM. The final feature Z_l is the output of the lth stage. The detailed structures of these three modules are shown in Fig. 2(c)–(e).

B. Multiscale Feature Extraction Module

As is widely known, feature extraction is one of the most fundamental processes in model training and inference, directly impacting prediction performance. Most mainstream object detectors are composed of three parts: the backbone, neck, and head. Among these, the backbone is primarily responsible for feature extraction, making the choice of structure crucial. Another important consideration is the variation in object scales. For example, in the MS COCO 2017 [23] dataset, object areas can range from less than 1,024 to over 9,216 pixels. This variation is even more pronounced in tiny-object detection datasets, as illustrated in Fig. 3. Despite the average scale of most objects being very small, there are also a significant number of objects with medium and large scales, particularly in the VisDrone2019 [22] dataset, which contains both tiny and large objects in the same scene. Although there are many studies focused on feature fusion, most existing attention mechanisms for object detection rely on a series of convolutional layers with fixed receptive fields to extract spatial or channel features. Some attention mechanism methods, such as the SKNet [19] and MRF [33], utilize multiscale features by incorporating convolutional layers with different kernel sizes, but this approach will introduce additional space and computational costs during training and inference.

To address the aforementioned issues, we design a novel MFEM. The structure of the MFEM is depicted in Fig. 2(c), which consists of four parallel average pooling layers with kernel sizes equal to 1, 3, 5, and 7, respectively, to extract features from different receptive fields. This operation is illustrated in the following equation:

$$M_l^k = \underset{k\times k}{\text{AvgPool}}\left(X_l\right), \qquad k = 1, 3, 5, 7 \tag{1}$$

where $\underset{k\times k}{\text{AvgPool}}$ represents the average pooling layer with kernel size k. X_l represents the feature map from the basic block in the lth stage of the backbone, and M_l^k represents the output feature map from the average pooling layer in the lth stage with kernel size k in the MFEM.

Unlike most existing methods, we use average pooling layers to extract multiscale information, replacing convolutional layers. As a result, our proposed MFEM module not only fully leverages multiscale features but also reduces the number of parameters compared to other multiscale feature extraction methods.

C. Gaussian Attention Module

The attention mechanism is an important method for enhancing the performance of object detectors, which draws inspiration from the human visual system. Most of the existing attention mechanisms for general object detection belong to channel attention and spatial attention. Channel attention mechanisms typically use global average pooling and a series of linear layers to extract channel information, which is then reshaped into a channel attention vector. Spatial attention, on the other hand, often employs two or more convolutional layers to determine the importance of different locations within a feature map. In the field of tiny-object detection, attention mechanisms are also essential, though there are two unique characteristics of tiny objects that existing methods may not fully consider. First, tiny objects may tend to be concentrated in a specific region of an image, as illustrated by an example from the AI-TOD dataset in Fig. 4(a). Additionally, as shown in Fig. 4(b), there is some noisy information surrounding the region, which can disrupt the detectors during training and inference.

To further enhance the performance of tiny-object detection, we propose an efficient and effective attention mechanism based on Gaussian distributions. We name it GAM. The proposed GAM consists of two parts: the Gaussian channel attention mechanism and the Gaussian spatial attention mechanism. For Gaussian channel attention, we first apply a global average pooling layer and a global maximum pooling layer to obtain the channel weight vector C_l^k, which reflects the importance of each channel. We then model this vector using a 1-D Gaussian distribution $N_c(\mu_c, \sigma_c^2)$, where the point with maximum weight x_c, is set as mean, and the square of half the channel length L_c, is used as variance, as shown in the

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following equation:

$$\mu_c = x_c, \qquad \sigma_c^2 = \frac{1}{4}L_c^2. \tag{2}$$

Following this, we can get the modeled Gaussian channel attention vector $G_{l,c}^k$, the weight of each channel is equal to the probability of each point in $N_c(\mu_c, \sigma_c^2)$, as shown in the following equation:

$$p_c\left(x;\mu_c,\sigma_c^2\right) = \frac{1}{\sqrt{2\pi}\sigma_c}\exp\left(-\frac{1}{2\sigma_c^2}(x-\mu_c)^2\right). \tag{3}$$

After that, we multiply the original feature map M_l^k and the modeled Gaussian channel attention vector $G_{l,c}^k$ to obtain the modeled Gaussian channel attention feature map $A_{l,c}^k$. The process is shown in the left part of Fig. 2(e).

Similarly, the process flow for Gaussian spatial attention is depicted on the right part of Fig. 2(e). To emphasize the important region, we model the spatial weight vector S_l^k using a 2-D Gaussian distribution $N_s(\mu_s, \Sigma_s)$, with the mean and covariance metrics defined in the following equation:

$$\mu_s = \begin{bmatrix} x_s \\ y_s \end{bmatrix}, \ \Sigma_s = \begin{bmatrix} \frac{1}{4}w_s^2 & 0 \\ 0 & \frac{1}{4}h_s^2 \end{bmatrix} \tag{4}$$

where (x_s, y_s) represents the point with maximum weight within the spatial weight vector, while w_s and h_s denote the width and height of the vector, respectively.

Similar to (3), we can obtain the modeled Gaussian spatial attention vector $G_{l,s}^k$ and get the weight of each point in it, which is shown in the following equation:

$$p_s\left(x;\mu_s,\Sigma_s\right) = \frac{1}{2\pi|\Sigma_s|^{1/2}}\exp\left(-\frac{1}{2}(x-\mu_s)^T\Sigma_s^{-1}(x-\mu_s)\right). \tag{5}$$

Then, we get the modeled Gaussian spatial attention feature map $A_{l,s}^k$ and add $A_{l,c}^k$ and $A_{l,s}^k$ together to obtain the modeled feature map A_l^k as the final output for GAM in the lth stage with kernel size k. The comparison between Fig. 4(b) and (c) illustrates the effectiveness of our GAM.

D. Dynamic Weighting Module

Multiscale feature fusion is a fundamental operation in object detection, but some existing methods simply combine these features, which can lead to information interference and hinder the ability to highlight valuable features. Considering that features from different receptive fields may contribute differently, we introduce a DWM, which can automatically assign appropriate weights to the output features from different receptive fields, as illustrated in Fig. 2(d). The weight assigned to the feature from the lth stage with kernel size k is denoted as α_l^k, as shown in the following equation:

$$\alpha_l^k = \frac{\mathrm{d}_1\left(A_l^k, X_l\right)}{w_l \times h_l}, \qquad k = 1,3,5,7 \tag{6}$$

where X_l represents the original feature map from the basic block in the lth stage and A_l^k represents the feature map after processing by the GAM. The term d_1 denotes the $L1$ distance between two metrics, while w_l and h_l refer to the width and height of the feature map from the lth stage, respectively.

When calculating the weights, we divide the $L1$ distance between two feature maps by their area, which ensures the weights fall within a range of zero to one, effectively functioning as a normalization operation.

To ensure the model fully utilizes attention information from various receptive fields, the weights of the attention features, α_l^k, increase with the difference between the attention feature map A_l^k and original feature map X_l. Additionally, we incorporate a residual connection to prevent information loss. The final output feature map for lth stage, Z_l, is shown in the following equation:

$$Z_l = X_l + \sum_{k=1,3,5,7} \alpha_l^k \times A_l^k. \tag{7}$$

E. Apply to Base Backbones

Our proposed MGAM, which integrates the MFEM, GAM, and DWM, can be seamlessly applied to most mainstream object detection backbones, such as the ResNet [6], VGG [46], MobileNet [47], and EfficientNet [48]. As depicted in Fig. 2(b), the MGAM can be easily incorporated into each stage of a backbone. Within each MGAM, we first employ the MFEM to capture features across different receptive fields. Then, we compute the channel and spatial attention for these features using the GAM. Finally, we assign varying weights to attention feature maps from different scales based on the difference between the original feature map from the basic block and the attention feature maps.

Importantly, we do not employ convolutional or linear layers during the feature extraction and attention mechanism processes. Although our MGAM introduces some additional computations, it has almost no impact on the GFLOPs and model size due to its low complexity, only slightly reducing the training speed of the model. Moreover, the pretrained weights for the backbones can be directly utilized.

IV. Experiments

A. Datasets and Evaluation Metrics

To comprehensively evaluate the effectiveness of our proposed method in remote sensing object detection and other related fields, we employ four types of datasets in experiments, including two satellite remote sensing datasets AI-TOD [45] and AI-TODv2 [67], one UAV remote sensing dataset VisDrone2019 [22], two road traffic domain datasets SODA-D [24] and TT100K [68], and one general-object detection dataset MS COCO 2017 [23].

1) AI-TOD and AI-TODv2 Datasets: AI-TOD [45] is a challenging benchmark dataset designed for optical remote sensing object detection tasks. It contains 28 036 images and 70 0621 object instances across eight categories. A distinguishing characteristic of this dataset is the extremely small size of objects, with an average size of just 12.8 pixels, and the largest object measuring less than 64 pixels. AI-TODv2 [67] is an enhanced version of the AI-TOD dataset, featuring the same set of images but with improvements such as an increased number of annotated instances, more realistic category distributions, and smaller average object sizes.

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2) VisDrone2019 Dataset: VisDrone2019 [22] is a large-scale benchmark for tiny-object detection, collected for a competition organized by Tianjin University. It includes 10 209 static images and 261 908 frames across ten categories, captured by UAVs. This dataset is particularly challenging due to the wide variations in object scales, as well as differences in height, angle, and lighting conditions across various scenarios.

3) SODA-D Dataset: SODA-D [24] is the latest published tiny-object detection dataset, which contains 24 828 high-quality images and 278 433 instances with precise annotations. Compared to existing traffic-oriented object detection datasets, SODA-D demonstrates superior characteristics in three aspects: broader category diversity, extremely small object sizes, and more comprehensive scene variations. It is also a challenging benchmark for tiny-object detection tasks.

4) TT100K Dataset: TT100K [68] is a large-scale traffic surveillance dataset jointly released by Tsinghua University and Tencent, focusing on traffic sign detection. It is widely used for research in the fields of autonomous driving and computer vision. It contains approximately 100 000 high-resolution labeled images covering 45 common categories of traffic signs as well as various scenes such as different weather, lighting, angles, and so on. Due to the small size of traffic signs, TT100K is also a classic benchmark for tiny-object detection tasks.

5) MS COCO 2017 Dataset: MS COCO 2017 [23] is one of the most widely used large-scale datasets for general-object detection. There are more than 118 K training images across eighty categories. It consists of the most common indoor, outdoor, natural landscapes, and urban street scenes in daily life, among other diverse environments.

6) Evaluation Metric: To facilitate comparison with existing methods, we use the general COCO metric [23] to compare the detection performance between existing methods and our proposed MGAM. The AP represents the mean average precision with the IoU increasing from 0.5 to 0.95 with a step of 0.05. The $AP_{0.5}$ and $AP_{0.75}$ represent the average precision when IoU equals 0.5 and 0.75. The subscripts of AP_{vt}, AP_t, AP_s, AP_m, and AP_l represent objects with the size of very tiny, tiny, small, medium, and large, respectively.

B. Implementation Details

We conduct our experiments using the widely adopted object detection frameworks MMDetection [74] and PyTorch [75]. During the training phase, for CNN-based methods, we use stochastic gradient descent (SGD) as the optimizer, with an initial learning rate set to 0.005, a momentum of 0.9, and a weight decay of 0.0001. The backbones of all models are pretrained on the ImageNet, and the batch size is set to 2 for each GPU. For the VisDrone2019, SODA-D, TT100K, and MS COCO 2017 datasets, following [24], the total number of training epochs is 12, with the learning rate decaying by a factor of 0.1 at the 8th and 11th epochs. For the AI-TOD and AI-TODv2 datasets, to fairly compare with existing state-of-the-art methods, following previous research [28], the total number of training epochs is set to 24, with the 0.1 factor decay occurring at the 16th and 22nd epochs. Other settings, such as data preprocessor and anchor generator, are consistent with those used in default of MMDetection [74]. The experiments are conducted on a computer equipped with an Intel[1] Xeon[1] Gold 6326 CPU and an NVIDIA GeForce RTX A6000 GPU. The versions of MMDetection, PyTorch, Python, and CUDA used are equal to 3.1.0, 2.0.1, 3.9.19, and 12.2, respectively. Considering that some methods only release results on certain datasets, the following results of these methods may not cover all datasets.

C. Results

1) Results on Satellite Remote Sensing Datasets: We conduct comprehensive comparisons between general object detectors, state-of-the-art methods, and our proposed MGAM. In Table I, the first five rows present results from two-stage anchor-based detectors, the sixth and seventh rows show results from one-stage anchor-based detectors, the next three and four rows display results from anchor-free and Transformer-based detectors, respectively. The following ten rows report results from state-of-the-art competitors on AI-TOD, and the next six rows present results from general object detectors embedded with our proposed MGAM. After embedding our mechanism into RetinaNet, FCOS, Faster R-CNN, Cascade R-CNN, and DetectoRS, our method achieves AP improvements of 2.2, 5.1, 9.5, 8.6, and 10.2 points, respectively. Compared with the latest published method with the best performance on AI-TOD, as shown in the 24th row in Table I, our method achieves significant improvements on AP and AP_{vt}. To further prove the effectiveness of our mechanism, as shown in the last row of Table I, we combine our mechanism with the state-of-the-art method, except embedding the MGAM into the backbone, all other structures are following the same setting, there is 1.1 points AP improvement which indicates our MGAM has strong universality. Some representative detection results before and after embedding our attention mechanism are illustrated in Fig. 5. The comparison between Fig. 5(a) and (b) demonstrates that our proposed MGAM accurately detects objects in both dense and sparse distributions, such as vehicles in the parking lot and ships near the harbor.

The detection accuracy on AI-TODv2 is presented in Table II. Similar to the experiments on AI-TOD, we compare the detection performance of our mechanism with several classic two-stage anchor-based detectors (shown in the first to fourth rows), one-stage anchor-based detectors (shown in the fifth to seventh rows), anchor-free detectors (shown in the eighth to 11th rows), and tiny-object detectors with state-of-the-art results (shown in the 12th and 16th rows). From Table II, it is obvious that our MGAM can also perform well for different types of object detectors on the AI-TODv2 dataset. Especially, embedding our proposed MGAM into the well-performing general object detector DetectoRS results in a 9.2 points AP improvement. Similar to the AI-TOD dataset, the accuracy improvements for very tiny and tiny objects are even more pronounced. Specifically, for DetectoRS, our mechanism achieves improvements of 11.4 and 13.2 points for AP_{vt} and AP_t, respectively. Compared with the latest published

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TABLE I
MAIN RESULTS ON AI-TOD. ALL MODELS ARE TRAINED ON THE TRAINVAL SET AND TESTED ON THE TEST SET

Method	Publication	Architecture	Backbone	AP↑	$AP_{0.5}$↑	$AP_{0.75}$↑	AP_{vt}↑	AP_t↑	AP_s↑	AP_m
two-stage anchor-based:										
TridentNet [49]	ICCV 2019	CNN-based	ResNet-50	7.5	20.9	3.6	1.0	5.8	12.6	14.0
Faster R-CNN [50]	TPAMI 2017	CNN-based	ResNet-50	11.1	26.3	7.6	0.0	7.2	23.3	33.6
Cascade RPN [51]	NeurIPS 2019	CNN-based	ResNet-50	13.3	33.5	7.8	3.9	12.9	18.1	26.3
Cascade R-CNN [52]	CVPR 2018	CNN-based	ResNet-50	13.8	30.8	10.5	0.0	10.6	25.5	36.6
DetectoRS [53]	CVPR 2021	CNN-based	ResNet-50	14.8	32.8	1.4	0.0	10.8	18.3	**38.0**
one-stage anchor-based:										
SSD-512 [54]	ECCV 2016	CNN-based	ResNet-50	7.0	21.7	2.8	1.0	4.7	11.5	13.5
RetinaNet [55]	ICCV 2017	CNN-based	ResNet-50	8.7	22.3	4.8	2.4	8.9	12.2	16.0
anchor-free:										
RepPoints [56]	ICCV 2019	CNN-based	ResNet-50	9.2	23.6	5.3	2.5	9.2	12.9	14.4
FCOS [57]	ICCV 2019	CNN-based	ResNet-50	12.6	30.4	8.1	2.3	12.2	17.2	25.0
M-CenterNet [45]	ICPR 2021	CNN-based	DLA-34	14.5	40.7	6.4	6.1	15.0	19.4	20.4
Transformer-based:										
DETR [58]	ECCV 2020	Transformer-based	ResNet-50	2.7	10.3	0.7	0.7	2.1	3.0	12.4
Deformable DETR [59]	ICLR 2021	Transformer-based	ResNet-50	9.9	28.0	-	-	-	-	-
DINO [60]	Arxiv 2022	Transformer-based	ResNet-50	13.2	34.4	8.1	2.7	7.3	11.9	27.5
Swin Transformer [61]	ICCV 2021	Transformer-based	-	13.8	32.7	8.3	3.1	8.6	14.8	28.3
state-of-the-art competitors:										
DetectoRS w/ CBAM [21]	ECCV 2018	CNN-based	ResNet-50	19.2	36.4	8.2	3.3	14.8	19.6	26.4
DetectoRS w/ ECA-Net [18]	CVPR 2020	CNN-based	ResNet-50	20.8	40.7	9.8	6.7	14.6	21.1	23.2
DetectoRS w/ SENet [17]	TPAMI 2020	CNN-based	ResNet-50	21.1	41.3	9.7	7.4	15.7	21.4	23.6
FSANet [62]	TGRS 2022	CNN-based	ResNet-50	20.3	48.1	14.0	6.3	19.0	26.8	36.7
HANet [43]	TCSVT 2024	CNN-based	ResNet-50	22.1	53.7	14.4	10.9	22.2	27.3	36.8
Crater-DETR [63]	TGRS 2024	Transformer-based	ResNet-50	18.1	42.1	-	-	-	19.7	-
KLDet [64]	TGRS 2024	CNN-based	ResNet-50	19.6	-	-	8.4	20.6	22.7	26.4
MENet [65]	TGRS 2024	CNN-based	Swin-T	23.2	56.2	15.0	9.7	23.9	25.3	34.4
YOLO-World [66]	CVPR 2024	Hybrid	-	18.5	38.1	12.4	6.8	13.6	19.3	31.7
TTFNet w/ GA [28]	TGRS 2024	CNN-based	MiT-B1	24.2	53.9	19.4	10.4	25.2	24.5	34.1
ours:										
RetinaNet w/ MGAM	-	CNN-based	ResNet-50	10.9	28.6	6.4	3.6	9.4	14.5	26.8
FCOS w/ MGAM	-	CNN-based	ResNet-50	17.7	43.8	13.0	3.4	12.8	19.1	35.5
Faster R-CNN w/ MGAM	-	CNN-based	ResNet-50	20.6	49.0	13.8	10.6	21.9	21.7	30.3
Cascade R-CNN w/ MGAM	-	CNN-based	ResNet-50	22.4	49.6	16.7	10.9	23.7	24.2	33.1
DetectoRS w/ MGAM	-	CNN-based	ResNet-50	25.0	53.2	19.3	**14.0**	25.3	**28.2**	36.8
TTFNet w/ GA + MGAM	-	CNN-based	MiT-B1	**25.3**	**54.8**	**19.9**	11.6	**26.0**	26.4	35.3

state-of-the-art DETR-like method on AI-TODv2, our MGAM also has 0.9 points of AP improvement.

2) Results on the UAV Remote Sensing Dataset: The detection performance of our proposed mechanism on the VisDrone2019 dataset is also highly competitive, as shown in Table III. The first five rows reflect results using official network structures and default settings, while the last five rows show results after embedding our MGAM into backbones. Compared with the baseline results for the RetinaNet, FCOS, Faster R-CNN, Cascade R-CNN, and DetectoRS, our method achieves AP improvements of 1.5, 5.1, 4.3, 5.1, and 3.0 points, respectively. Special attention should be given to the improvements in detection performance for very tiny (AP_{vt}) and tiny (AP_t) objects, as shown in the fourth and fifth columns of Table III. In the case of Faster R-CNN, the baseline AP_{vt} is only 0.1 points, indicating that most very tiny objects are not well detected. However, after embedding MGAM into the backbone, this accuracy improves by 7.2 points. The significant improvements in AP_{vt} and AP_t clearly demonstrate that our proposed method is well-suited for tiny-object detection tasks, especially scenes containing very tiny objects. In addition, we also compare our method with some of the latest proposed state-of-the-art detectors, as shown in the sixth to ninth rows; our MGAM still has competitive detection performance.

3) Results on Tiny-Object Detection Datasets in Other Fields: To further validate the effectiveness of our mechanism, we conduct a series of experiments on two tiny-object detection datasets in the traffic field, SODA-D and TT100K. The detection results for the latest tiny-object detection dataset SODA-D are shown in Table IV. The first nine rows list classic general object detectors chosen as baselines, while the last four rows show the detection results after embedding the MGAM into these detectors. The tenth to fourteenth rows display some latest published state-of-the-art methods on the SODA-D dataset. For RetinaNet, FCOS, Faster R-CNN, and Cascade R-CNN, our mechanism brings AP improvements of 0.6, 1.3, 0.7, and 0.5 points, respectively. Some detection examples for each category are displayed in Fig. 6.

The experimental results on the traffic surveillance dataset TT100K are shown in Table V. We compare the detection performance of some baseline detectors (as shown in the first to ninth rows) and some state-of-the-art detectors specifically designed for traffic signs (as shown in the tenth and 11th

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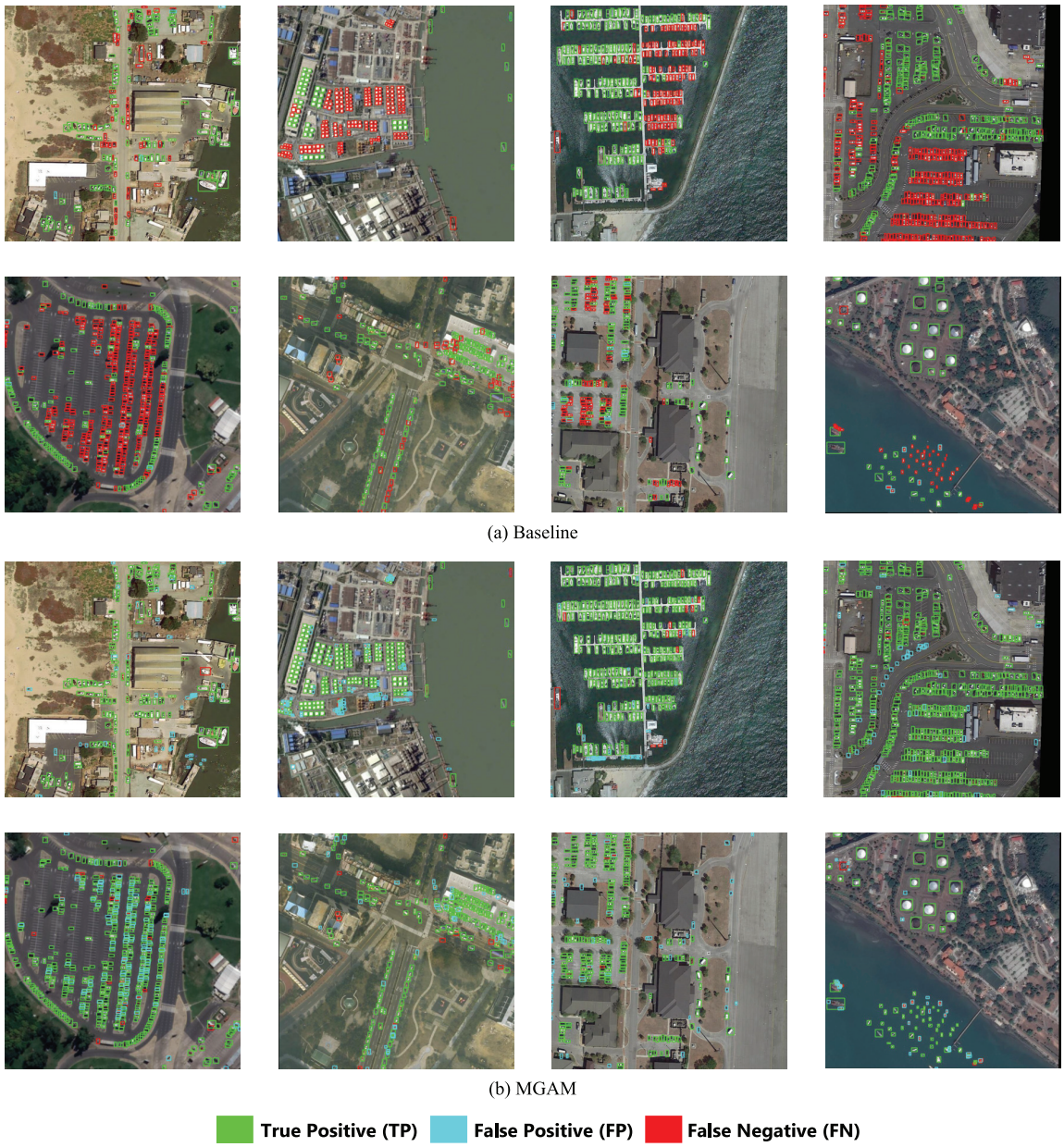


Fig. 5. Comparison between before and after using our MGAM on the AI-TOD test set based on Faster R-CNN. (a) Original detection performance while (b) embedded with our proposed mechanism. The green, blue, and red rectangles represent true positive (TP), false positive (FP), and false negative (FN), respectively. If the IoU between prediction and ground truth is larger than 0.5, the prediction is regarded as correct. The performance improvements brought by our method are obvious.

rows) on the TT100K dataset. For each detector, we report the AP, $AP_{0.5}$, and $AP_{0.75}$, respectively. On the basis of the baseline detectors RetinaNet, FCOS, Faster R-CNN, Cascade R-CNN, and DetectoRS, by applying our proposed MGAM, AP improvements of 0.3, 0.2, 2.7, 2.4, and 2.0 can be achieved, respectively. Compared to the state-of-the-art method OpenTransMind, our method can also bring a 2.0 $AP_{0.5}$ performance improvement. The experimental results demonstrate that our proposed MGAM is not only suitable for tiny-object detection in remote sensing images but also has excellent performance in other related fields.

4) Results on the General-Object Detection Dataset: Our method not only excels in tiny-object detection datasets but also demonstrates excellent performance on general object detection tasks. As shown in Table VI, the first row presents results using the official ResNet-50 backbone, while the second row shows results using ResNet-50 embedded with our MGAM. Our method achieves a 1.4-point AP improvement compared with the baseline, highlighting the strong robustness and adaptability of our method for objects of various sizes.

D. Ablation Study

1) Effectiveness of Each Module: Our proposed MGAM consists of three modules. To clearly illustrate the contribution of each module, we conduct a set of ablation studies on the AI-TOD dataset. As shown in Table VII, all models are based on Faster R-CNN, trained on the AI-TOD trainval set, and tested on the AI-TOD test set.

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TABLE II
MAIN RESULTS ON AI-TODV2. ALL OF THE SETTINGS ARE AS SAME AS THOSE FOR AI-TOD

Method	Publication	Architecture	Backbone	AP↑	$AP_{0.5}$↑	$AP_{0.75}$↑	AP_{vt}↑	AP_t↑	AP_s↑	AP_m↑
two-stage anchor-based:										
TridentNet [49]	ICCV 2019	CNN-based	ResNet-50	10.1	24.5	6.7	0.1	6.3	19.8	31.9
Faster R-CNN [50]	TPAMI 2017	CNN-based	ResNet-50	12.8	29.9	9.4	0.0	9.2	24.6	37.0
Cascade R-CNN [52]	NeurIPS 2019	CNN-based	ResNet-50	15.1	34.2	11.2	0.1	11.5	26.7	38.5
DetectoRS [53]	CVPR 2021	CNN-based	ResNet-50	16.1	35.5	12.5	0.1	12.6	28.3	40.0
one-stage anchor-based:										
YOLOv3 [69]	ArXiv 2018	CNN-based	DarkNet-53	4.1	14.6	0.9	1.1	4.8	7.7	8.0
RetinaNet [55]	ICCV 2017	CNN-based	ResNet-50	8.9	24.2	4.6	2.7	8.4	13.1	20.4
SSD-512 [54]	ECCV 2016	CNN-based	VGG-16	10.7	32.5	4.0	2.0	8.7	16.8	28.0
anchor-free:										
RepPoints [56]	ICCV 2019	CNN-based	ResNet-50	9.3	23.6	5.4	2.8	10.0	12.3	18.9
FoveaBox [70]	TIP 2020	CNN-based	ResNet-50	11.3	28.1	7.4	1.4	8.6	17.8	32.2
FCOS [57]	ICCV 2019	CNN-based	ResNet-50	12.0	30.2	7.3	2.2	11.1	16.6	26.9
Grid R-CNN [71]	CVPR 2019	CNN-based	ResNet-50	14.3	31.1	11.0	0.1	11.0	25.7	36.7
state-of-the-art competitors:										
ESG_TODNet [37]	GRSL 2024	CNN-based	ResNet-50	24.6	55.1	18.1	9.5	24.0	29.4	35.6
ORFENet [44]	TGRS 2024	CNN-based	ResNet-50	24.8	55.4	18.2	9.7	24.4	28.7	35.1
FFCA-YOLO [72]	TGRS 2024	CNN-based	DarkNet-53	27.7	61.7	-	-	-	-	-
EDADet [73]	TGRS 2024	Transformer-based	-	29.9	67.2	24.2	12.3	30.1	36.1	42.0
DQ-DETR [11]	ECCV 2024	Transformer-based	ResNet-50	30.2	68.6	22.3	15.3	30.5	36.5	44.6
ours:										
RetinaNet w/ MGAM	-	CNN-based	ResNet-50	12.5	33.1	7.2	3.0	9.9	17.4	31.8
FCOS w/ MGAM	-	CNN-based	ResNet-50	18.2	46.4	12.7	3.7	13.6	20.5	36.4
Faster R-CNN w/ MGAM	-	CNN-based	ResNet-50	21.5	52.0	14.0	9.1	21.9	23.6	33.4
Cascade R-CNN w/ MGAM	-	CNN-based	ResNet-50	23.5	52.7	17.5	9.1	23.9	27.1	35.7
DetectoRS w/ MGAM	-	CNN-based	ResNet-50	25.3	55.3	19.0	11.5	25.8	28.0	37.5
DQ-DETR w/ MGAM	-	Transformer-based	ResNet-50	**31.1**	**69.8**	**22.9**	**15.7**	**31.3**	**37.6**	**46.1**

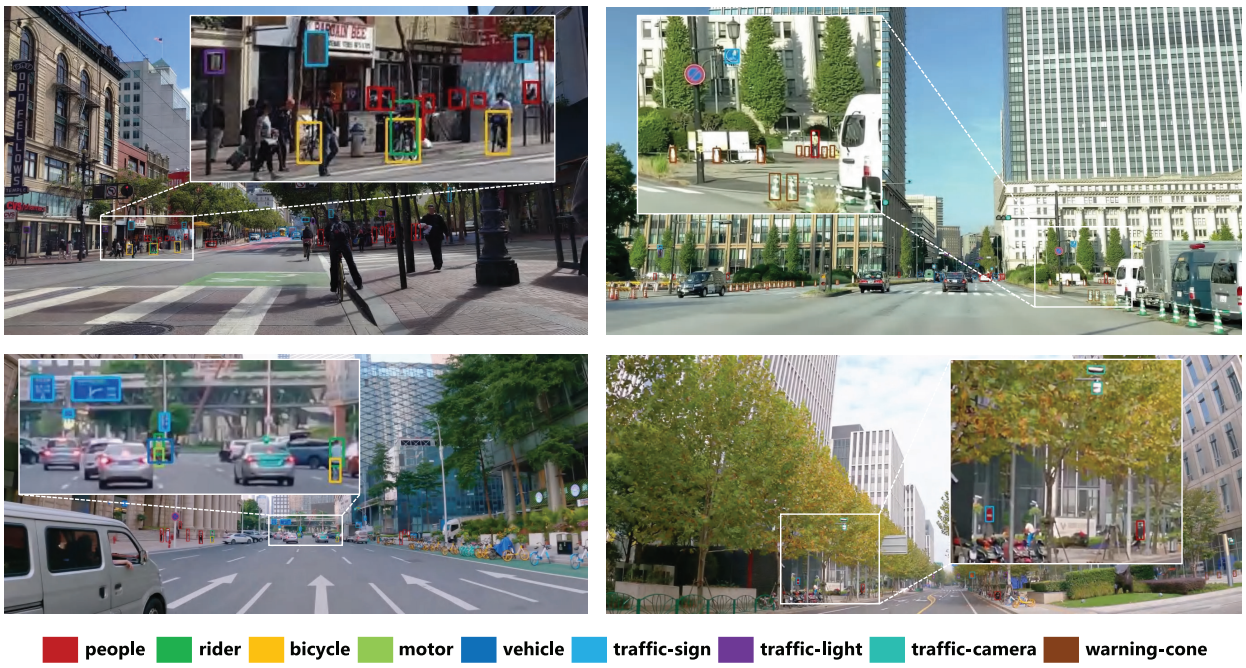


Fig. 6. Visual detection results on the test set of the SODA-D dataset using Faster R-CNN embedded with our proposed MGAM. The rectangles with different colors, respectively, represent different categories of objects.

MFEM aims to obtain multiscale feature maps from the original output feature. As shown in Table VII, the first row presents the original detection results using official Faster R-CNN, while the second row shows the detection results after adding MFEM into each stage of the backbone. The addition of the MFEM results in a 6.4 points AP improvement, underscoring the effectiveness of fusing feature maps from different receptive fields, which is crucial given the significant variations in object scales within tiny-object detection datasets.

The GAM is used to compute attention vectors, which can reflect the importance of each channel and location, and then make the model pay more attention to the important channel

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TABLE III

MAIN RESULTS ON VISDRONE2019. ALL MODELS ARE TRAINED ON THE TRAINING SET AND TESTED ON THE VAL SET

Method	AP↑	$AP_{0.5}$↑	AP_{vt}↑	AP_t↑
baseline:				
RetinaNet [55]	7.9	15.9	0.1	0.7
FCOS [57]	14.1	25.5	0.1	2.1
Faster R-CNN [50]	22.3	38.0	0.1	6.2
Cascade R-CNN [52]	22.5	38.5	0.5	6.8
DetectoRS [53]	25.7	41.7	0.5	7.6
state-of-the-art competitors:				
RFLA [76]	27.4	45.3	4.5	12.9
YOLOv8s [77]	27.5	46.8	-	-
YOLOv10s [78]	24.7	54.7	7.6	24.7
LIS-DETR [79]	23.8	40.1	-	-
ours:				
RetinaNet w/ MGAM	9.4	17.4	0.3	2.1
FCOS w/ MGAM	19.2	37.1	5.6	11.4
Faster R-CNN w/ MGAM	26.6	48.7	7.3	16.0
Cascade R-CNN w/ MGAM	27.6	48.5	7.3	16.3
DetectoRS w/ MGAM	**28.7**	**50.4**	**8.0**	**16.8**

TABLE IV

MAIN RESULTS ON SODA-D.ALL MODELS ARE TRAINED ON THE TRAINING SET AND TESTED ON THE TEST SET

Method	AP↑	$AP_{0.5}$	$AP_{0.75}$↑
baseline:			
Faster R-CNN [50]	28.9	59.7	24.2
Cascade R-CNN [52]	31.2	59.9	27.8
RetinaNet [55]	28.2	57.6	23.7
CornerNet [80]	24.6	49.5	21.7
CenterNet [81]	21.5	48.8	15.6
FCOS [57]	23.9	49.5	19.9
RepPoints [56]	28.0	55.6	24.7
Sparse RCNN [82]	24.2	50.3	20.3
YOLOX [83]	26.7	53.4	23.0
state-of-the-art competitors:			
RFLA [76]	29.7	60.2	25.2
CenterNet-Elite [84]	24.8	51.9	19.3
KLDet [64]	25.9	53.8	21.4
DILA [85]	30.4	61.4	26.1
Libra-SOD [86]	30.9	58.3	27.9
ours:			
RetinaNet w/ MGAM	28.8	56.1	24.8
FCOS w/ MGAM	25.2	50.5	22.1
Faster R-CNN w/ MGAM	29.6	57.9	26.4
Cascade R-CNN w/ MGAM	**31.7**	58.1	**29.2**

TABLE V

MAIN RESULTS ON TT100K. ALL MODELS ARE TRAINED ON THE TRAINING SET AND TESTED ON THE TEST SET

Method	AP↑	$AP_{0.5}$↑	$AP_{0.75}$↑
baseline:			
RetinaNet [55]	21.6	30.6	25.3
FCOS [57]	50.2	65.8	58.0
SSD-512 [54]	-	68.7	-
Mask R-CNN [87]	-	70.8	-
ScratchDet [88]	-	74.0	-
Sparse R-CNN [82]	53.2	69.5	62.8
Faster R-CNN [50]	58.9	74.1	69.3
Cascade R-CNN [52]	63.5	80.1	74.8
DetectoRS [53]	64.2	82.9	75.7
state-of-the-art competitors:			
CAB Net [89]	-	78.0	-
Open-TransMind [90]	-	83.0	-
ours:			
RetinaNet w/ MGAM	21.9	31.1	25.7
FCOS w/ MGAM	50.4	66.0	58.2
Faster R-CNN w/ MGAM	61.6	80.9	72.8
Cascade R-CNN w/ MGAM	65.9	84.4	77.0
DetectoRS w/ MGAM	**66.2**	**85.0**	**78.2**

TABLE VI

COMPARISON ON MS COCO 2017. ALL MODELS ARE BASED ON FASTER R-CNN WITH RESNET-50 BACKBONE, TRAINED ON THE TRAINVAL35K SET, AND TESTED ON THE MINIVAL SET

Method	AP	AP_s	AP_m	AP_l
ResNet-50 (baseline)	36.5	21.0	39.8	46.8
MGAM (ours)	**37.9**	**21.8**	**42.2**	**48.7**

TABLE VII

ABLATION STUDY OF THE PROPOSED THREE MODULES. ALL MODELS ARE BASED ON THE FASTER R-CNN AND AI-TOD DATASETS

MFEM	GAM	DWM	AP	$AP_{0.5}$	$AP_{0.75}$
			11.1	26.3	7.6
✓			17.5	45.7	11.7
✓	✓		19.7	48.5	13.6
✓	✓	✓	**20.6**	**49.0**	**13.8**

and region, in the meantime, suppress surrounding noisy information. As shown in the third row of Table VII, based on the MFEM, incorporating the GAM into the backbone yields an additional 2.2 points AP improvement, which indicates the effectiveness of the GAM. More importantly, different from most of the existing attention mechanisms, our GAM uses pooling layers to replace convolutional or linear layers, thereby not introducing any parameters to the network.

As shown in the fourth row of Table VII, based on the MFEM and GAM, the DWM can also bring 0.9 points AP improvement to the model, which indicates that assigning different weights to the feature maps with different receptive fields is effective. In order to provide a deeper understanding of the DWM, we also analyze the relationship between the weight distribution of feature maps with different receptive fields during the training process and the average size of objects in each image. As shown in Fig. 7, it can be found that as the average size of objects in each image increases, the weight difference between feature maps with different receptive fields gradually decreases. This indicates that smaller objects are more sensitive to feature maps with different receptive fields and also proves that multiscale feature fusion can provide richer information for tiny objects. Our proposed DWM assigns larger weights to feature maps with lower similarity to the input feature, further improving the information richness of the fused feature maps. When all three proposed modules are used simultaneously, the detection accuracy increases to 20.6, which is 9.5 points higher than the original accuracy, clearly demonstrating the effectiveness of our proposed three modules.

2) Performance of Different Modeling Methods in the Attention Mechanism: To evaluate the impact of different attention modeling methods on detection performance, we only modify the modeling method of the GAM proposed in Section III-C

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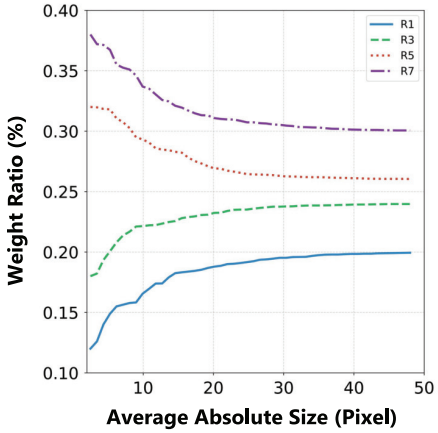


Fig. 7. Relationship between the weight ratio distribution of feature maps with different receptive fields and the average absolute size of objects in each image. $R1$, $R3$, $R5$, and $R7$ represent the feature maps obtained through the average pooling layers and GAM, where the kernel sizes of the average pooling layers are equal to 1×1, 3×3, 5×5, and 7×7, respectively.

while keeping its structure unchanged. Specifically, we replace the Gaussian attention modeling method with Laplacian attention modeling, Exponential attention modeling, and Learnable soft attention modeling, respectively. Then, we start from a mathematical perspective and define an indicator named the target-to-background ratio (TBR) to reflect the feature extraction performance. For each feature map $X \in \mathbb{R}^{C\times H\times W}$, where C, H, and W represent its number of channels, height, and width, respectively. The TBR of the feature map is the ratio of the signal values of the area where objects are located (valuable area) to the signal values of the area where no object is located (background area). The definition is as follows:

$$\mathrm{TBR} = \frac{\sum_{i=0}^{H}\sum_{j=0}^{W}\sum_{k=0}^{C}\mathbb{1}_{i,j}^{\mathrm{obj}}X(k,i,j)}{\sum_{i=0}^{H}\sum_{j=0}^{W}\sum_{k=0}^{C}\mathbb{1}_{i,j}^{\mathrm{noobj}}X(k,i,j)} \tag{8}$$

where $\mathbb{1}_{i,j}^{\mathrm{obj}}$ and $\mathbb{1}_{i,j}^{\mathrm{noobj}}$ are symbols to indicate whether the position (i, j) on the feature map is the location of objects. If there are objects at the location (i, j), $\mathbb{1}_{i,j}^{\mathrm{obj}}$ is equal to 1 and $\mathbb{1}_{i,j}^{\mathrm{noobj}}$ is equal to 0. If there is no object at the location (i, j), $\mathbb{1}_{i,j}^{\mathrm{obj}}$ is equal to 0 and $\mathbb{1}_{i,j}^{\mathrm{noobj}}$ is equal to 1.

From a mathematical theory perspective, we take the 1-D Gaussian distribution and 1-D Exponential distribution as examples. We assume that the number of channels is $2L$ and the objects are symmetrically concentrated in the center position, with this position as the origin of the coordinate axis. Therefore, we only need to consider the part of $x > 0$. Assuming that the objects are distributed in the region of $0 <= x < r$, this region is the signal region, and the region of $r <= x < L$ is the background region. We can use the integration of Gaussian distribution probability density function and the Exponential distribution probability density function to represent the sum of signal values of target and background modeled using the two probability distributions in the regions of $0 <= x < r$ and $r <= x < L$, and their ratio is the TBR under the two distributions, respectively. As shown in (9) and (10), TBR_G and TBR_E represent the TBR under Gaussian and Exponential distributions, respectively,

$$\mathrm{TBR}_G = \frac{\int_0^r f_G(x)\,dx}{\int_r^L f_G(x)\,dx} \tag{9}$$

$$\mathrm{TBR}_E = \frac{\int_0^r f_E(x)\,dx}{\int_r^L f_E(x)\,dx} \tag{10}$$

where $f_G(x) = (1/\sigma(2\pi)^{1/2})e^{-((x-\mu)^2/2\sigma^2)}$ and $f_E(x) = \lambda e^{-\lambda x}$ are the probability density functions of Gaussian distribution and Exponential distribution, respectively. In $f_G(x)$, since we assume that objects are located in the center position and model the entire channel, $\mu = 0$ and $\sigma = L$. For λ in (10), we assume that it is also modeling the entire channel similar to a Gaussian distribution; therefore, we assume there may be an equal value for two distributions when $x = L$, which satisfies the relationship in the following equation:

$$\frac{1}{L\sqrt{2\pi}}e^{-\frac{L^2}{2L^2}} = \lambda e^{-\lambda L} \tag{11}$$

where, after simplification, we can obtain the following equation:

$$\lambda L e^{-\lambda L} = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}} \tag{12}$$

where assuming $\lambda L = k$, then (12) becomes $ke^{-k} = (1/(2\pi)^{1/2})e^{-(1/2)} \approx 0.242$, therefore, k is approximately equal to 0.5. For convenience of calculation, in the process of modeling the Exponential distribution, we take $\lambda = (1/2L)$. To prove the relationship between TBR_G and TBR_E, we construct an auxiliary function $R(x)$, as shown in the following equation:

$$R(x) = \frac{f_G(x)}{f_E(x)} = \frac{2}{\sqrt{2\pi}}e^{-\frac{x^2}{2L^2}+\frac{x}{2L}} \tag{13}$$

where let $(d/\mathrm{dx})R(x) = 0$, we can obtain $x = (1/2)L$. Therefore, $R(x)$ monotonically increases at $x \in [0, (1/2)L)$, monotonically decreases at $x \in [(1/2)L, L)$, and $R(L) = (2/(2\pi)^{1/2}) < 1$. According to the integral ratio theorem, we can obtain the relationship between the integrals of two distributions, as shown in the following equation:

$$\frac{\int_0^r f_G(x)\,dx}{\int_0^r f_E(x)\,dx} > \frac{\int_r^L f_G(x)\,dx}{\int_r^L f_E(x)\,dx} \tag{14}$$

where equivalent to the following equation:

$$\frac{\int_0^r f_G(x)\,dx}{\int_r^L f_G(x)\,dx} > \frac{\int_0^r f_E(x)\,dx}{\int_r^L f_E(x)\,dx}. \tag{15}$$

Therefore, we can prove that $\mathrm{TBR}_G > \mathrm{TBR}_E$. To illustrate this intuitively, we can transform (9) and (10) into expressions only for L and r, we take $0 < L <= 10$, $0 <= r < L$, and plot the relationship between L, r, and the TBR of the corresponding Gaussian and exponential distributions. This conclusion can also be observed from Fig. 8(a).

For Laplacian distribution $f_L(x) = (1/2b)e^{-(|x-\mu|/b)}$, we still assume that the objects are distributed at the center position and only consider the case where $x > 0$, so we set $\mu = 0$. To make the Laplacian distribution satisfy the condition that when $x = L$, the Gaussian distribution, exponential distribution, and Laplacian distribution have the same values as much as possible, we set $b = 2L$ in the modeling process. At

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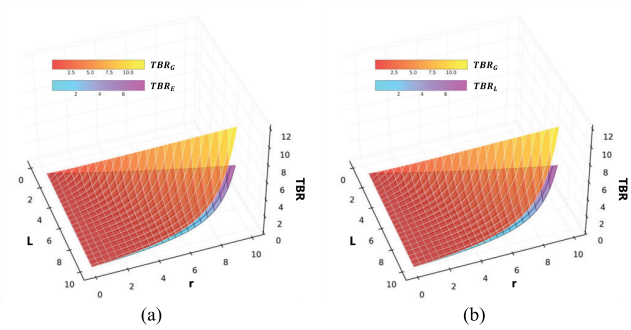


Fig. 8. Visual comparison of the relationship of L, r, and the TBR. (a) Relationship between Gaussian distribution (TBR_G) and exponential distribution (TBR_E). (b) Relationship between Gaussian distribution (TBR_G) and Laplacian distribution (TBR_L).

TABLE VIII

COMPARISON BETWEEN DIFFERENT MODELING METHODS IN THE ATTENTION MECHANISM. ALL MODELS ARE BASED ON THE FASTER R-CNN AND AI-TOD DATASETS

Modeling Method	TBR	AP	$AP_{0.5}$	$AP_{0.75}$
Baseline	0.051	11.1	26.3	7.6
Laplacian attention	0.101	16.4	43.7	10.1
Exponential attention	0.105	16.5	43.9	10.0
Learnable soft attention	0.053	13.1	36.8	6.9
Gaussian attention (ours)	**0.124**	**20.6**	**49.0**	**13.8**

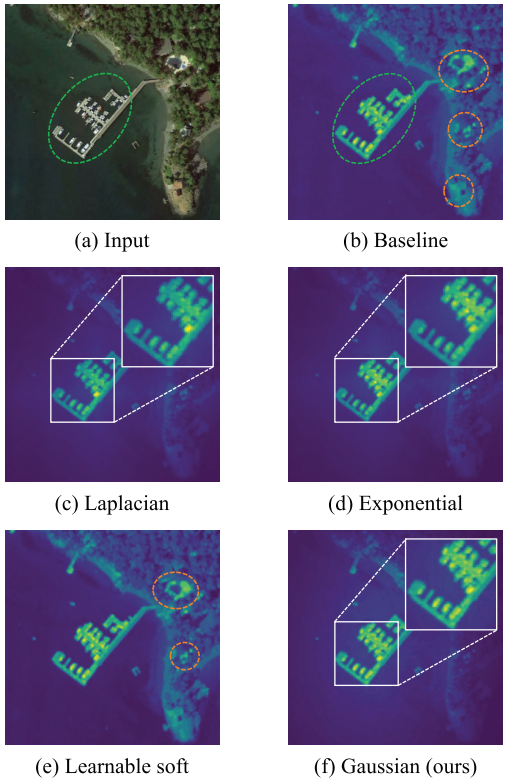


Fig. 9. Visual comparisons of the performance of different modeling methods in the attention mechanism. (a) Original input image, (b) feature map without using the attention mechanism, and (c)–(f) feature maps using different modeling methods in the attention mechanism. In each feature map, the green circle represents the region where most of the objects appear, and the orange circle represents the region that contains some noisy information.

this time, the Laplacian distribution is $f_L(x) = (1/4L)e^{-(x/2L)}$. Similar to the method in (13), we can also prove that the TBR of the Gaussian distribution is larger than that of the Laplacian distribution. Through observation, it can be seen that the difference between the Laplacian distribution and the exponential distribution is only a fixed multiple, so the TBR of the Laplacian and exponential distributions is theoretically identical. The comparison between Gaussian distribution (TBR_G) and Laplacian distribution (TBR_L) is shown in Fig. 8(b). For learnable soft attention, since it does not have a specific equation, we compare and analyze it with our proposed Gaussian attention from an experimental perspective in the following part.

We also calculate the average TBR of feature maps generated using different attention mechanism modeling methods during a training epoch, as shown in Table VIII. The higher the TBR value, the higher the weight of the object's region in the feature map; at the same time, the more obvious the noise suppression effect in the background region, which indicates that the attention modeling method is more effective. Besides, we also conduct experimental evaluation under the same configuration. The results are shown in Table VIII. We can find that although other attention modeling methods have a certain improvement in accuracy compared to the baseline result, our proposed Gaussian attention modeling still has significant advantages. Fig. 9 shows a visual comparison of feature maps using different attention modeling methods. Although Laplacian modeling and exponential modeling also emphasize the central region where the objects are located, they do not highlight the peripheral objects well. The learnable soft attention modeling method does not effectively utilize the distribution characteristics, resulting in poor suppression of background noise. Overall, our proposed Gaussian attention mechanism not only emphasizes the region where objects are located, but also effectively suppresses background noise, which is consistent with the comparison results in Table VIII.

3) Analysis of Time and Space Cost During Inference and Training: To comprehensively evaluate our proposed method, we compare our MGAM with some typical baseline methods on AI-TOD dataset in terms of speed (per image), GFLOPs, and parameters during inference, as well as speed (per image), model size, GPU MEM (GPU memory usage), and CPU MEM (CPU memory usage) during training. As shown in Table IX, the baseline methods include CNN-based methods, Transformer-based methods, and attention-based methods. All experiments are conducted on an NVIDIA GeForce RTX A6000 GPU with the same settings in above mentioned implementation details. When calculating speed and GFLOPs, the input size of the model is 800×800, where the AP, AP_{vt}, and AP_t are obtained from training on the AI-TOD trainval set and testing on the AI-TOD test set. In Table IX, from the comparison between our MGAM and the corresponding CNN-based baseline methods, it can be found that our proposed MGAM does not increase the number of parameters while significantly improving detection accuracy. Due to its relatively simple operations, the increase in GFLOPs and model size can be ignored, and only a small amount of time cost and GPU/CPU memory usage will be increased. For some Transformer-based methods, as shown in the fourth to sixth rows in Table IX, although the time and space cost during

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TABLE IX
COMPARISON OF TIME AND SPACE COST DURING INFERENCE AND TRAINING BETWEEN OUR PROPOSED MGAM AND SOME BASELINE METHODS

Method	AP	AP_{vt}	AP_t	Inference			Training			
				Speed (ms)	GFLOPs	Parameters (MB)	Speed (ms)	Model size (MB)	GPU MEM (GB)	CPU MEM (GB)
CNN-based baseline:										
RetinaNet [55]	8.7	2.4	8.9	10	129.8	36.5	60	292.0	6.2	3.0
FCOS [57]	12.6	2.3	12.2	10	123.3	32.1	60	257.5	2.6	3.0
DetectoRS [53]	14.8	0.0	10.8	35	243.3	54.4	220	989.5	11.6	3.2
Transformer-based baseline:										
DETR [58]	2.7	0.7	2.1	10	60.5	41.6	60	333.4	3.3	3.1
DINO [60]	13.0	2.7	7.3	25	179.0	47.6	140	382.7	10.3	3.1
Swin Transformer [61]	13.5	3.1	8.6	15	85.4	38.4	90	308.9	7.9	3.1
Attention-based baseline:										
DetectoRS w/ CBAM [21]	19.2	3.3	14.8	40	243.4	59.9	225	1,034.2	11.8	3.9
DetectoRS w/ ECA-Net [18]	20.8	6.7	14.6	35	243.3	54.4	220	989.5	11.7	3.9
DetectoRS w/ SENet [17]	21.2	7.4	15.7	40	243.3	59.9	225	1,034.1	11.9	3.9
ours:										
RetinaNet w/ MGAM	10.9	3.6	9.4	15	129.8	36.5	80	292.0	6.4	3.0
FCOS w/ MGAM	17.7	3.4	12.8	15	123.3	32.1	80	257.5	2.9	3.0
DetectoRS w/ MGAM	25.0	14.0	25.3	40	243.3	54.4	235	989.5	11.8	3.4

TABLE X
LATENCY AND FPS OF THE MGAM ON EDGE COMPUTING DEVICES

Method	Latency (ms)		FPS	
	iPhone 12	Pixel 6	iPhone 12	Pixel 6
RetinaNet w/ MGAM	3.3	217.4	303.0	4.6
FCOS w/ MGAM	3.1	208.3	322.6	4.8
DetectoRS w/ MGAM	8.3	541.6	120.5	1.8

inference and training may be reduced in some cases, their detection accuracy is often not particularly ideal. Compared to the classic attention-based methods CBAM and SENet, our MGAM has achieved a significant reduction in parameters and model size, as well as a noticeable improvement in accuracy (especially for AP_{vt} and AP_t), with only an increase of about 5% in training time. Compared to the ECA-Net, our method also brings a significant improvement in accuracy. In summary, our MGAM has achieved satisfactory performance in terms of detection accuracy and inference, and training costs.

4) Performance on Edge Computing Devices: To evaluate the performance of our methods on edge computing devices, we deploy and test our MGAM on edge computing devices. Specifically, we first convert the trained detectors into models that can be run on Apple and Google mobile devices through the compilation tools CoreMLTools and XNNPACK, respectively. Then, we test them on iPhone 12 (Neural Engine) and Pixel 6 (CPU), respectively. For each model and device, we report two indicators, latency and FPS. During the test processing, the input size is equal to 800×800 and the batch size is equal to 1. In addition, we conduct 10 epochs of warmup before each testing, and the reported results are average values of 1000 epochs, ensuring the error ranges within 0.1 ms. As shown in Table X, the widely used one-stage detectors RetinaNet and FCOS, as well as the two-stage detector DetectoRS, are utilized to deploy and test. Benefit from the acceleration capability of the Neural Engine in iPhone 12, our MGAM can complete high-speed real-time inference on it. For the DetectoRS model, which has the best detection performance but a large size, the latency and FPS on iPhone 12 still achieve satisfactory results. For the CPU-only device, Pixel 6, its inference speed is also in a usable state. Since our MGAM does not introduce too much computational complexity, it can be well deployed on edge computing devices.

E. Discussion

Significant improvements are evident after applying our proposed MGAM, as reflected in the results from Tables I–VII, particularly in the accuracy gains for very tiny and tiny objects. The main enhancements of our method can be summarized in three main points.

First, our method effectively detects objects of various sizes, ranging from very tiny to large. This capability is primarily due to our MFEM, which fully leverages information from different receptive fields. Unlike existing methods, our module does not introduce any additional parameters to the network, as it uses pooling layers instead of convolutional layers.

Second, as shown in the comparison between the first and second rows of Fig. 10, our attention mechanism effectively suppresses noisy information from regions without objects. Additionally, regions surrounding objects are valuable for detection, as they may contain contextual information. The importance of these regions tends to decrease as the distance increases, a phenomenon akin to the characteristic of the Gaussian distribution. From the second row in Fig. 10, it is evident that our attention mechanism not only directs the focus of the model to regions where objects are more likely to appear but also preserves contextual information. This capability is one of the main reasons why our method performs well on tiny-object detection datasets.

Third, acknowledging that different receptive fields may hold varying levels of importance within a given scene, we design a DWM to significantly reduce interference from irrelevant information. Importantly, our method is highly robust across different networks and depths, and it does not require any additional parameters for training. This makes it both effective and efficient, with broad applicability across various use cases.

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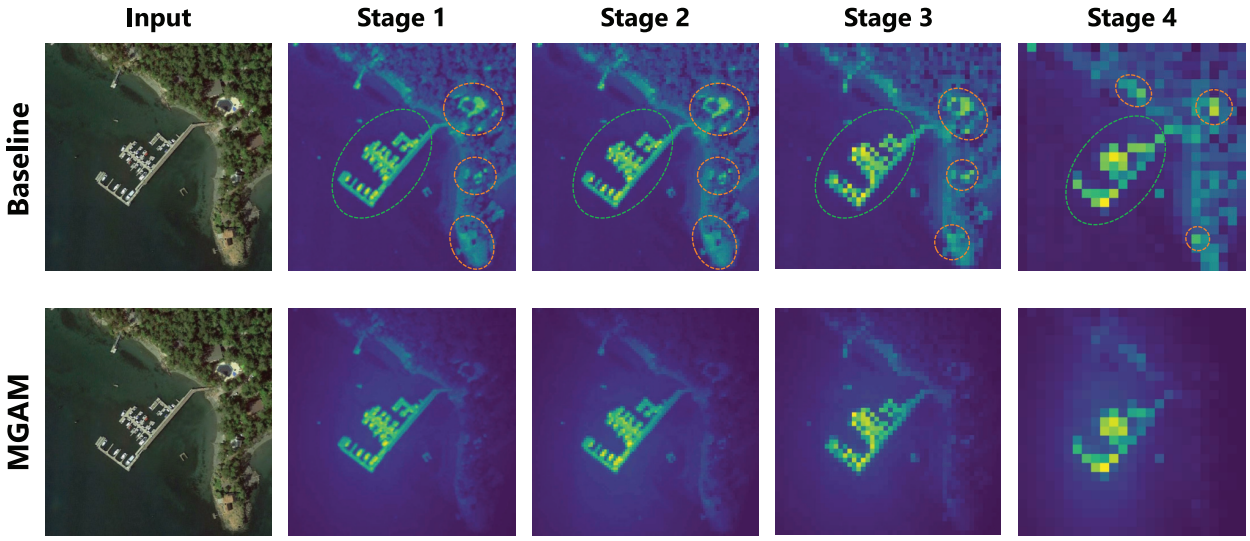


Fig. 10. Visual comparisons of the feature extraction performance in each stage of the backbone. In each feature map, the green circle represents the region where most of the objects appear, and the orange circle represents the region that contains some noisy information.

V. Conclusion

In this article, we first provide a brief review of existing methods for tiny-object detection, attention mechanisms, and feature fusion. Considering the characteristics of tiny objects, we then propose a novel MGAM that incorporates three modules. It not only integrates both channel and spatial attention mechanisms, but also dynamically utilizes information from multiscale receptive fields without adding extra parameters. Subsequently, we conduct a series of experiments on five classic datasets, with extensive results demonstrating the effectiveness and robustness of our method.

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